MODULE 4: SPECTROSCOPIC & DIFFRACTION TECHNIQUES

Official FAT Study Textbook & Step-by-Step Question Bank Solutions (CO: 3)

Aligned with Official VIT BACHY101 Syllabus, Department Question Bank, and Past FAT Papers

Scope of Module 4 (Official VIT Syllabus - All 8 Topics):

(1) Electromagnetic radiation and light-matter interactions, (2) UV-Visible spectroscopy and Beer-Lambert law, (3) Electronic transitions and selection rules, (4) Chromophores and auxochromes, (5) Absorption shifts: Bathochromic, hypsochromic, hyperchromic, and hypochromic, (6) Powder X-ray diffraction (XRD) and Bragg's law, (7) Debye-Scherrer equation for crystallite sizing, and (8) Worked Question Bank Problems.

1. Core Syllabus Definitions & Theoretical Foundations (Line-by-Line)

1. Electromagnetic Radiation & Light-Matter Interactions

Electromagnetic radiation propagates as transverse oscillating electric (E) and magnetic (B) fields at the speed of light ($c = 3.0 \times 10^8$ m/s). Photons carry discrete quantized energy $E = h\nu = \frac{hc}{\lambda}$. When interacting with molecular matter:

- UV-Visible radiation ($200 - 800$ nm) promotes valence electrons between bonding and antibonding molecular orbitals.
- Infrared radiation ($2.5 - 15 \mu$m) excites molecular bond stretching and bending vibrations.
- X-Rays ($0.01 - 10$ nm) eject inner core electrons and undergo constructive elastic diffraction from crystal planes.

2. UV-Visible Spectroscopy

An optical absorption spectroscopy technique measuring the attenuation of monochromatic ultraviolet or visible light transmitted through a chemical solution as a function of wavelength, providing qualitative structural identity and quantitative concentration determination.

3. Beer-Lambert Law

The fundamental quantitative relationship governing optical absorption in homogeneous solutions:

$$A = \log_{10}\left(\frac{I_0}{I}\right) = \varepsilon c l$$

where A is the dimensionless optical absorbance (optical density), I_0 is the incident light beam intensity, I is the transmitted intensity, c is the molar concentration of the absorbing solute in mol/L, l is the optical path length of the sample cuvette (standard quartz cuvette $l = 1.0$ cm), and ε is the molar absorptivity (molar extinction coefficient) in $\text{L} \cdot \text{mol}^{-1} \cdot \text{cm}^{-1}$.

4. Electronic Transitions in Molecules

The quantum mechanical excitation of an electron from an occupied molecular orbital into an unoccupied antibonding orbital upon photon absorption:

- $\sigma \rightarrow \sigma^*$: Requires high energy vacuum UV photons (< 180 nm), observed exclusively in saturated hydrocarbons (e.g. $C-C, C-H$ single bonds in alkanes).
- $n \rightarrow \sigma^*$: Requires moderate energy UV ($150-250$ nm), observed in saturated compounds bearing non-bonding heteroatom lone pairs (e.g. $C-O, C-N, C-Cl$).
- $\pi \rightarrow \pi^*$: Strong, symmetry-allowed transition ($200-400$ nm) exhibiting high molar absorptivity ($\varepsilon > 10000$), characteristic of unsaturated double bonds, alkenes, carbonyls, and conjugated polyenes.
- $n \rightarrow \pi^*$: Symmetry-forbidden transition ($250-400$ nm) with low molar absorptivity ($\varepsilon < 100$), occurring at longer wavelengths in carbonyls and nitrogen heterocycles.

5. Chromophore

An unsaturated covalent functional group containing mobile π-electrons that absorbs electromagnetic radiation in the UV or visible region ($200-800$ nm), imparting color or UV absorption bands to the molecule (e.g. $-C=C-, -C\equiv C-, -C=O, -N=N-, -NO_2$, and aromatic phenyl rings).

6. Auxochrome

A saturated substituent group carrying non-bonding lone pairs (n-electrons) that does not absorb UV radiation on its own, but when bonded directly to a chromophore, extends orbital conjugation, shifts the absorption maximum to longer wavelengths, and enhances the absorption intensity (e.g. $-OH, -NH_2, -OCH_3, -SH, -Cl$).

7. Bathochromic Shift (Red Shift)

The displacement of an absorption band maximum (λ_{max}) toward longer wavelengths (lower photon energy). Caused by extended π-electron conjugation, auxochromic substitution, or changes in solvent polarity.

8. Hypsochromic Shift (Blue Shift)

The displacement of an absorption band maximum (λ_{max}) toward shorter wavelengths (higher photon energy). Caused by loss of orbital conjugation (e.g. protonation of aniline $-NH_2$ to anilinium $-NH_3^+$) or specific solvent hydrogen-bonding interactions with lone pairs.

9. Hyperchromic & Hypochromic Effects

- **Hyperchromic Effect:** An increase in absorption intensity (molar extinction coefficient ε_{max} increases).
- **Hypochromic Effect:** A decrease in absorption intensity (molar extinction coefficient ε_{max} decreases).

10. Powder X-Ray Diffraction (Powder XRD)

A non-destructive physical analytical technique for identifying crystallographic phases, unit cell dimensions, and sub-micrometer crystallite domain sizes in polycrystalline materials. Monochromatic $Cu\ K\alpha$ X-rays ($\lambda = 1.5406$ Å) incident on randomly oriented crystallites generate concentric diffraction cones that satisfy Bragg's condition.

11. Bragg's Law of Diffraction
The geometric and physical condition for constructive interference of monochromatic X-rays scattered from parallel crystallographic planes separated by interplanar spacing d:

$$2d \sin\theta = n\lambda$$

where d is interplanar lattice spacing, θ is the glancing Bragg angle of incidence, n is an integer representing the diffraction order ($n = 1, 2, 3...$), and λ is the X-ray wavelength (1.5406 Å for $Cu\ K\alpha$).

12. Debye-Scherrer Equation for Crystallite Size
The fundamental formula relating diffraction peak broadening to the mean coherent crystallite domain size D:

$$D = \frac{K\lambda}{\beta \cos\theta}$$

where D is average crystallite size in nanometers, K is the dimensionless Scherrer shape factor ($K \approx 0.9$ for spherical crystallites), λ is X-ray wavelength (0.15406 nm), β is the full width at half maximum (FWHM) of the diffraction peak expressed strictly in radians (corrected for instrumental broadening: $\beta = \sqrt{\beta_{\text{measured}}^2 - \beta_{\text{inst}}^2}$), and θ is the Bragg angle corresponding to the peak center.

2. In-Depth Chemical Theory & Step-by-Step Derivations

Derivation of the Beer-Lambert Law

Consider a parallel beam of monochromatic light of incident intensity I_0 entering an absorbing homogeneous solution of molar concentration c. Let I be the intensity after traversing a differential thickness dx of the solution. The fractional decrease in light intensity ($-d\frac{I}{I}$) across the thin layer dx is directly proportional to the number of absorbing molecules in that layer:

$$-\frac{dI}{I} = k'c dx$$

where k' is a proportionality absorption constant. Integrating across the entire cuvette length from $x = 0$ ($I = I_0$) to $x = l$ ($I = I_t$):

$$-\int_{I_0}^{I_t} \frac{dI}{I} = k'c \int_0^l dx \Rightarrow -\ln\left(\frac{I_t}{I_0}\right) = \ln\left(\frac{I_0}{I_t}\right) = k'cl$$

Converting the natural logarithm to base 10 ($\log_{10}(z) = \frac{\ln(z)}{2.3026}$):

$$\log_{10}\left(\frac{I_0}{I_t}\right) = \left(\frac{k'}{2.3026}\right)cl = \varepsilon cl$$

where $\varepsilon = \frac{k'}{2.3026}$ is the molar absorptivity (extinction coefficient). Defining Absorbance $A = \log_{10}\left(\frac{I_0}{I_t}\right)$:

$$\boldsymbol{A = \varepsilon cl}$$

3. VIT FAT Exam Blueprint & Master Formula Sheet

Technique / Law	Governing Equation / Formula	VIT Exam Key Insight

Beer-Lambert Law	$A = \log_{10}\left(\frac{I_0}{I}\right) = -\log_{10}(T) = \varepsilon c l$	Transmittance $T = \frac{I}{I_0}$; $\%T = 100 \times 10^{-A}$.
Photon Energy Conversion	$E = h\nu = \frac{hc}{\lambda} = \frac{1240\ \text{eV·nm}}{\lambda}$	$h = 6.626 \times 10^{-34}$ J · s; $c = 3.0 \times 10^8$ m/s.
Bragg's Diffraction Law	$2d \sin\theta = n\lambda \Rightarrow d = \frac{n\lambda}{2\sin\theta}$	For cubic crystal: $d_{hkl} = \frac{a}{\sqrt{h^2+k^2+l^2}}$.
Debye-Scherrer Formula	$D = \frac{K\lambda}{\beta \cos\theta}$	β (FWHM) MUST be converted from degrees to radians!
FWHM Degree to Radian	$\beta_{\text{rad}} = \beta_{\text{deg}} \times \left(\frac{\pi}{180}\right)$	Instrumental correction: $\beta = \sqrt{\beta_{\text{obs}}^2 - \beta_{\text{inst}}^2}$.

4. Worked VIT Question Bank & Past FAT Numerical Problems

VIT FAT Problem 1 (10 Marks): Beer-Lambert Quantitative Analysis

A solution of a compound has an optical absorbance of 0.699 at $\lambda_{\text{max}} = 450$ nm when placed in a 1.00 cm path length cuvette. Given its molar absorptivity is $\varepsilon = 1.40 \times 10^4$ L · mol^{-1} · cm^{-1}.
(a) Calculate the molar concentration (c) of the compound in the solution.
(b) Calculate the percentage transmittance ($\%T$) of this solution.
(c) If the solution is diluted to three times its original volume ($3V_0$), what will be its new absorbance and percentage transmittance?

Step-by-Step Solution:

- **Part (a): Molar Concentration:**

$$c = \frac{A}{\varepsilon l} = \frac{0.699}{1.40 \times 10^4\ \text{L} \cdot \text{mol}^{-1} \cdot \text{cm}^{-1} \times 1.00\ \text{cm}} = \mathbf{4.993 \times 10^{-5}\ mol/L}$$

- **Part (b): Percentage Transmittance ($\%T$):**

$$A = -\log_{10}(T) \Rightarrow T = 10^{-A} = 10^{-0.699} = 0.19999$$

$$\%T = T \times 100 = \mathbf{20.0\%}$$

- **Part (c): 3-Fold Dilution:** When volume is tripled, concentration becomes $c' = \frac{c}{3}$. Since absorbance is directly proportional to concentration:

$$A' = \frac{A}{3} = \frac{0.699}{3} = \mathbf{0.233}$$

New transmittance:

$$T' = 10^{-0.233} = 0.5848 \Rightarrow \%T' = \mathbf{58.48\%}$$

Boxed Final Answers:
(a) Molar Concentration $c = \mathbf{4.99 \times 10^{-5}}$ **M**
(b) Percentage Transmittance $\%T = \mathbf{20.0\%}$
(c) Diluted Absorbance $A' = \mathbf{0.233}$ and $\%T' = \mathbf{58.5\%}$

VIT FAT Problem 2 (10 Marks): Powder XRD & Debye-Scherrer Crystallite Sizing

In a Powder XRD experiment using $Cu\ K\alpha$ radiation ($\lambda = 1.5406$ Å $= 0.15406$ nm), a prominent diffraction peak appears at $2\theta = 38.40\,^\circ$. The measured Full Width at Half Maximum (FWHM) of this peak is $\beta = 0.45\,^\circ$.
(a) Calculate the interplanar lattice spacing d for this crystal plane.
(b) Using the Debye-Scherrer equation with shape factor $K = 0.90$, calculate the average crystallite size (D) in nanometers.

Step-by-Step Solution:

- **Part (a): Bragg Interplanar Spacing (d):** Glancing angle θ:

$$\theta = \frac{2\theta}{2} = 38.40\,\frac{^\circ}{2} = 19.20\,^\circ$$

$$\sin(19.20\,^\circ) = 0.32887$$

Applying Bragg's law ($n = 1$):

$$d = \frac{\lambda}{2\sin\theta} = \frac{1.5406\text{ Å}}{2 \times 0.32887} = \frac{1.5406}{0.65774} = \mathbf{2.342\text{ Å}} \quad (\mathbf{0.2342\text{ nm}})$$

- **Part (b): Debye-Scherrer Crystallite Sizing:** Convert peak broadening β from degrees to radians:

$$\beta_{\text{rad}} = 0.45\,^\circ \times \left(\frac{\pi}{180}\right) = 0.45 \times 0.0174533 = \mathbf{7.854 \times 10^{-3}\text{ rad}}$$

$$\cos\theta = \cos(19.20\,^\circ) = 0.94438$$

Applying Debye-Scherrer formula:

$$D = \frac{K\lambda}{\beta\cos\theta} = \frac{0.90 \times 0.15406\text{ nm}}{7.854 \times 10^{-3} \times 0.94438}$$

$$D = \frac{0.138654}{7.4171 \times 10^{-3}} = \mathbf{18.69\text{ nm}}$$

Boxed Final Answers:
(a) Interplanar Lattice Spacing $d = \mathbf{2.342}$ Å $= \mathbf{0.2342}$ **nm**
(b) Average Crystallite Size $D = \mathbf{18.69}$ **nm**